MATH 110

Dani Nguyen

Spring 2026

Contents

1

1.1 Lecture 1 - Introduction to PDEs

What is a ODE? Recall that an ordinary differential equation (ODE) are equations for functions of a single variable, $u(t),\ t \in \mathbb{R}$

$$\frac{du}{dt} = u(1-u) \ \text{ is an ODE}$$

Here we call t the independent variable and u the state/dependent variable. In general, an ODE has the form $F(t) = (t, u, \frac{du}{dt}, \frac{d^2u}{dt^2}, \dots)$

Note that the state variable can have many components.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \in \mathbb{R}^2$$

then

$$\frac{du}{dt} = \mathbf{u} \;\text{ is an ODE}$$

What is a PDE? A PDE is an equation for a function with more than 1 independent variable, e.g. $u(x, y)$.

$$\begin{aligned}\text{Notation: } u_x &= \frac{\partial u}{\partial x} = \partial_x u \\ u_y &= \frac{\partial u}{\partial y} = \partial_y u \\ &\text{etc.}\end{aligned}$$

Examples of PDEs

- Heat/diffusion equation for $u(x, t)$

$$u_t = k u_{xx} \quad k \text{ constant}$$

- Wave equation for $u(x, t)$

$$u_{tt} = c^2 u_{xx} \quad c \text{ constant}$$

- Laplace's equation for $u(x, y)$

$$u_{xx} + u_{yy} = 0$$

$\boxed{\text{ex}}$ Consider the heat flow in a bar of length l

- $u(x, t)$ is the temperature of the bar at position x, time t
- Suppose the ends are held at temperature 0, and the initial temperature is $\phi(x)$
- The temperature evolves according to the heat equation $u_t = k u_{xx}$ where $t \geq 0, 0 \leq x \leq l$, k constant.

We also have:

- Boundary conditions (BCs):

$$u(0, t) = u(l, t) = 0 \quad \forall t$$

- Initial conditions (ICs):

$$u(x,0) = \phi(x)$$

Remarks

- The order of the PDE is the order of the highest derivative that occurs.
- The heat equation above is 2^{nd} order
- A general 2^{nd} order PDE for $u(x,t)$ is:

$$G(x,t,u,u_x,u_t,u_{xx},u_{tt},u_{xt}) = 0$$

- A PDE in the above form is linear if G is a linear function of u and all of its derivatives. It is nonlinear otherwise.
- A linear PDE is homogeneous if every term contains u or some derivative of u. It is nonhomogeneous/inhomogeneous otherwise.

$\boxed{\text{ex}}$ Consider the linearity and homogeneity of the following equations.

1. $u_t + e^{x^2 t} u_{xt} = 0$
   is a linear and homogeneous PDE.
2. $u_t + 3xu_{xx} = \frac{t}{x}$
   is a linear and inhomogeneous PDE because of the $\frac{t}{x}$ term.
3. $u_t + uu_x = 0$
   is a nonlinear PDE because of the uu_x term.
4. $u_{tt} + u_x + sin(u) = 0$
   is nonlinear because of the $sin(u)$ term.

Operating notation & linearity
We can write the heat equation $u_t - ku_{xx} = 0$ as $Lu = 0$ where

$$L = (\partial_t - k\partial_x^2)$$

We call L the differential operator. L acts on (smooth) functions $u(x,t)$ to produce a new function.

$$\begin{aligned} L(u) = Lu &= (\delta_t - k\partial_x^2)u \\ &= u_t - ku_{xx} \end{aligned}$$

L is a linear operator iff

$$\begin{aligned} L(u+v) &= Lu + Lv \\ &\qquad \forall \text{ functions } u, v \end{aligned}$$

and

$$\begin{aligned} L(cu) &= cLu \\ &\quad \forall \text{ constants } c \in \mathbb{R} \\ &\quad \forall \text{ functions } u \end{aligned}$$

$\boxed{\text{ex}}$ Check if the heat equation has a linear differential operator.

1.

$$\begin{aligned} L(u+v) &= (\partial_t + k\partial_x^2)(u+v) \\ &= u_t + ku_{xx} + v_t + kv_{xx} \\ &= Lu + Lv \quad \checkmark \end{aligned}$$

2.

$$\begin{aligned} L(cu) &= cu_t + cku_{xx} \\ &= cLu \text{ for constant } c \quad \checkmark \end{aligned}$$

Remarks

- If L is a linear operator then $Lu = f$ is a linear PDE

$$\underbrace{u_t + ku_{xx}}_{Lu} = \underbrace{e^{x^2 t}}_{f(x,t)} \text{ is linear}$$

- If L is a linear operator then the PDE $Lu = f$ is
  - homogeneous if $f = 0$
  - inhomogeneous if $f \neq 0$
- A linear, homogeneous PDE $Lu = 0$ has the superposition principle

- If u_1, u_2 are solutions to the PDE then $au_1 + bu_2$ is a solution for constants a, b.

$$\begin{aligned} Lu_1 &= Lu_2 = 0 \\ \Rightarrow L(au_1 + bu_2) &= 0 \end{aligned}$$

- The distinction between linear & nonlinear PDEs is very important.

<u>Solutions to PDEs</u>

- A solution to a PDE is a function u in which satisfies the PDE and any BCs/ICs upon direct substitution.
- The general solution to an ODE involves arbitrary constants. The general solution to a PDE involves arbitrary functions.
- Typically, we are interested in specific solutions.

$\boxed{\text{ex}}$ Consider the heat equation $u_t - u_{xx} = 0$.

$$\begin{aligned} u_1(x,t) &= x^2 + 2t \text{ is a soln} \\ \rightarrow \partial_t u_1 - \partial_x^2 u_1 &= 2 - 2 = 0 \\ u_2(x,t) &= e^{-t} \sin x \text{ is a soln} \\ \rightarrow \partial_t u_2 - \partial_x^2 u_2 &= -e^{-t} \sin x + e^{-t} \sin x = 0 \end{aligned}$$

$\boxed{\text{ex}}$ Consider the equation $u_x = tx$ (1st order, linear, inhomogeneous)

$$\begin{aligned} Lu = f &\text{ with } L = \partial_x \\ f(x,t) &= tx \end{aligned}$$

We can solve this by direct integration on both sides wrt x.

$$u(x,t) = \frac{1}{2}tx^2 + A(t)$$

Where $A(t)$ is an arbitrary function of t

$\boxed{\text{ex}}$ Consider the equation $u_{tt} - 4u = 0$ (2nd order, linear, homogeneous)

$$\begin{aligned} Lu &= f \\ \rightarrow L &= (\partial_t^2 - 4) \\ f(x,t) &= 0 \end{aligned}$$

There is no explicit x-dependence

- We can treat this as an ODE with x as a parameter.
- 'Constants' depend on x
- General solution:

$$u(x,t) = A(x)e^{-2t} + B(x)e^{2t}$$

1.2 Lecture 2 - Characteristics

Recap:

- Heat equation for $u(x,t)$

$$u_t - ku_{xx} = 0, \quad k > 0, \text{constant}$$

- We can write the heat equation as $Lu = 0$, where $L = (\partial_t - k\partial_x^2)$ is a differential operator.
- L is linear $\leftrightarrow$ $L(u+v) = Lu + Lv$ and $L(cu) = cLu$ $\forall$ functions u, v, constant c
- PDE: $Lu = f$ is linear if L is linear
- Linear PDE: $Lu = f$ is homogeneous if $f = 0$, inhomogeneous otherwise.

First-order linear PDE
Consider functions $u(x,y)$

$$u : \mathbb{R}^2 \to \mathbb{R}$$

Often we think of x, y as spatial variable and t as time variables. The simplest PDE we can have following this is:

$$u_x = 0$$

Which is 1st-order, linear, homogeneous. Also note that there is no explicit y-dependence. Therefore we can treat this like an ODE, and solve with direct integration of both sides.

$$u(x,y) = f(y)$$

where $f(y)$ is an arbitrary function of y.

Now consider $au_x + bu_y = 0$, which is a 1st-order, linear, homogeneous PDE with constants $a, b \in \mathbb{R}$.

$$\begin{aligned} \text{Let } V &= (a,b) \in \mathbb{R}^2 \\ \text{Recall } \nabla u &= (u_x, u_y) \\ \text{Then } V \cdot \nabla u &= (a,b) \cdot (u_x, u_y) \\ &= au_x + bu_y = 0 \end{aligned}$$

$\Rightarrow$ Directional derivative in direction V is 0.
$\Rightarrow$ u is constant on lines parallel to V.

Let $V^\perp = (b, -a)$, so that $V \cdot V^\perp = 0$.
Lines parallel to V satisfy

$$\begin{aligned} V^\perp \cdot (x,y) &= \text{constant} \\ \Rightarrow bx - ay &= \text{constant} \end{aligned}$$

Then u is constant on lines parallel to V.

$$\begin{aligned} \Rightarrow u(x,y) &= f(bx - ay) \\ \text{for an arbitrary function of } f \end{aligned}$$

<u>Check</u>:

$$\begin{aligned} &(a\partial_x + b\partial_y)\ f(bx - ay) \\ &= abf' - baf' = 0 \\ &\Rightarrow f' \text{ is a derivative of } f : \mathbb{R} \to \mathbb{R} \end{aligned}$$

Characteristic Coordinates. In this example, the lines parallel to V are called characteristics. Here u is constant along the characteristics.

We can write points (x, y) in characteristic coordinates (ξ, η). Where ξ is the x-intercept of the characteristic line, and η is the distance along that line to the point (x, y).

$$
\begin{aligned}
\text{So } x &= \xi + a\eta = x(\xi, \eta) \\
y &= b\eta = y(\xi, \eta) \\
\xi &= x - a\eta \\
\eta &= \frac{y}{b} \\
\xi &= x - \frac{a}{b}y \\
b\xi &= bx - ay
\end{aligned}
$$

Transform PDE to (ξ, η) coordinates.

Chain rule:

$$
\begin{aligned}
\frac{\partial u}{\partial \eta} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \eta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \eta} \\
\Rightarrow u_\eta &= au_x + bu_y = 0
\end{aligned}
$$

Using characteristic coordinates (ξ, η), we've transformed to the PDE $u_\eta = 0$ for $u(\xi, \eta)$.

<u>Solution</u>:

$$
\begin{aligned}
u &= f(\xi) \\
\text{where } \xi &= \frac{1}{b}(bx - ay) \\
\Rightarrow u &= g(bx - ay)
\end{aligned}
$$

Where g is an arbitrary function.

<u>Remarks</u>:

- η is coordinates along characteristics
- u is constant on characteristics $\Rightarrow u_\eta = 0$
- ξ labels different characteristics. There are other choices for ξ but any choice $\tilde{\xi}$ would have $\tilde{\xi} = h(\xi)$

Method of Characteristics

Method for solving PDEs of form

$$a(x,y)u_x + b(x,y)u_y = 0 \quad \text{for } u(x,y)$$

Idea: Let $A(x,y) = (a(x,y), b(x,y))$

The directional derivative $A \cdot \nabla u = 0$

$\Rightarrow u$ is constant along curves w/ slope A.

- These are characteristic curves.
- Find coord η along chars
- Find coord ξ which labels chars
- PDE transforms to $u_\eta = 0$, general soln: $u = f(\xi)$

$\boxed{\text{ex}}$ $u_x + yu_y = 0$

The directional derivative along $(1, y)$ are 0.

The chars $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = 1, \ \frac{dy}{d\eta} = y$$

At $\eta = 0$, take $(x(0), y(0)) = (0, \xi)$

Solve:

$$\begin{aligned}
&\frac{dx}{d\eta} = 1, \ x(0) = 0 \\
&\Rightarrow x = \eta \\
&\frac{dy}{d\eta} = y, \ y(0) = \xi \\
&\Rightarrow y = \xi e^\eta
\end{aligned}$$

This follows that $\xi = ye^{-x}$.

$$\begin{aligned}
\Rightarrow u_\eta &= u_x \frac{\partial x}{\partial \eta} + u_y \frac{\partial y}{\partial \eta} \\
&= u_x + \xi e^\eta u_y \\
&= u_x + yu_y = 0 \\
\Rightarrow u_\eta &= 0 \\
\Rightarrow u &= f(\xi) = f(ye^{-x}) \text{ with arbitrary function } f
\end{aligned}$$

General solution:
Suppose we are given $u(0,y) = y^2$

$$\Rightarrow \text{Specific solution}$$
$$u(0,y) = f(y) = y^2$$
$$\Rightarrow u(x,y) = (ye^{-x})^2$$

Alternatively, from parametric equations,

$$\frac{dx}{d\eta} = 1 \qquad \frac{dy}{d\eta} = y$$
$$\Rightarrow \frac{dy}{dx} = y$$
$$\Rightarrow y = Ce^x$$

$\Rightarrow u$ is constant along chars.
$\Rightarrow u(x, Ce^x) = u(0, C)$
$= u(0, ye^{-x})$
$= f(e^{-x}y)$ for arbitrary f

$\boxed{\text{ex}}$ $u_x + 2xy^2u_y = 0$
Characteristic curves: $(x(\eta), y(\eta))$

$$\frac{dx}{d\eta} = 1 \qquad \frac{dy}{d\eta} = 2xy^2$$
$$\frac{dy}{dx} = 2xy^2$$

$$\int \frac{1}{y^2}dy = \int 2x dx$$
$$-\frac{1}{y} = x^2 - c$$
$$y = \frac{1}{c - x^2}$$

u is constant along chars.
$\Rightarrow u(x,y) = f(c) = f(x^2 + \frac{1}{y})$ is the general soln.

Advection, transport

Consider $u(x,t)$ describing quantity u at position x, time t.

$$u_t + cu_x = 0 \quad (c > 0)$$

Chars: $(t(\eta), x(\eta))$ satisfy

$$\begin{aligned} &\frac{dt}{d\eta} = 1 \qquad \frac{dx}{d\eta} = c \\ &\Rightarrow \frac{dx}{dt} = c \\ &\Rightarrow x = ct + \xi \quad \text{via integration} \end{aligned}$$

General solution:

$$u = f(\xi) = f(x - ct)$$

1.3 Lecture 3 - Conservation Laws

Recap:

- Solve PDEs of the form $a(x,y)u_x + b(x,y)u_y = 0$
- The char curves $(x(\eta), y(\eta))$ satisfy

$$\frac{dx}{d\eta} = a(x,y) \qquad \frac{dy}{d\eta} = b(x,y)$$

- u is constant along chars.

$\boxed{\text{ex}}$ Given $\frac{1}{x^2}u_x + u_y = 0$

$$\frac{dx}{d\eta} = \frac{1}{x^2} \qquad \frac{dy}{d\eta} = 1$$

Method 1: divide to get parametric eq.

$$\begin{aligned} \frac{dy}{dx} &= x^2 \\ y &= \frac{1}{3}x^3 + \xi \end{aligned}$$

where constant ξ labels the char.

The general solution is $u = f(\xi) = f(y - \frac{1}{3}x^3)$ for an arbitrary function f. This can be confirmed by plugging u back into the PDE.

Method 2: solve parametric equations.

$$\begin{aligned} &\frac{dx}{d\eta} = \frac{1}{x^2} \qquad \frac{dy}{d\eta} = 1 \\ &\Rightarrow \int x^2 dx = d\eta \qquad y = \eta + B \\ &\Rightarrow \frac{1}{3}x^3 = \eta + A \qquad y = \eta + B \end{aligned}$$

We can absorb the two constants into one ξ

$$\begin{aligned} &\Rightarrow \frac{1}{3}x^3 = \eta + \xi \qquad y = \eta \\ &\Rightarrow y = \frac{1}{3}x^3 - \xi \end{aligned}$$

We should check that the chars cover the whole plane, i.e. no chars have been 'missed', via sketching or finding

$$\begin{aligned} \eta &= \eta(x, y) \\ \xi &= \xi(x, y) \end{aligned}$$

Coord transformation is also useful

$$\begin{matrix} \frac{1}{3}x^3 = \eta + \xi \\ y = \eta \end{matrix} \Rightarrow \begin{matrix} x = (3\eta + 3\xi)^{\frac{1}{3}} \\ y = \eta \end{matrix}$$

This transform our PDE

$$\begin{aligned} \frac{1}{x^2}u_x + u_y = 0 \;&\Leftrightarrow\; u_\eta = 0 \\ \text{Gen soln: } u &= f(\xi) \\ &= f(\frac{1}{3}x^3 - y) \end{aligned}$$

with arbitrary function f

$\boxed{\text{ex}}$ Advection
Consider function $u(x,t)$ describing a quantity u at position x and time t

$$u_t + cu_x = 0 \tag{Advection}$$

with constant $c > 0$
Chars $(t(\eta), x(\eta))$ satisfy

$$\frac{dt}{d\eta} = 1 \qquad \frac{dx}{d\eta} = c$$

Solve

$$\begin{aligned} t &= \eta + A \\ x &= c\eta + B \end{aligned}$$

Absorb into one constant using $A = 0, B = \xi$

$$\begin{aligned} \Rightarrow t &= \eta \\ x &= c\eta + \xi \\ \Rightarrow x &= ct + \xi \end{aligned}$$

By inverting, $\eta = t,\ \xi = x - ct$.
Now we have an transformed PDE

$$u_\eta = u_t + cu_x = 0$$

General solution is

$$\begin{aligned} u(x,t) &= f(\xi) \\ &= f(x - ct) \end{aligned}$$

for an arbitrary f.

Suppose we had initial condition $u(x,o) = \phi(x)$, then $u(x,0) = f(x) = \phi(x) \Rightarrow$ solution is $u(x,t) = \phi(a - ct)$.

<u>Remarks</u>:
We can solve more complicated PDEs using chars, for example,

$$u_t + cu_x = f(u)$$

Transform to coords (ξ, η) with $t = \eta$, $x = c\eta + \xi$

$$u\eta = f(u)$$

which can be solved as an ODE

Conservation Laws. (Logan 1.2)

- Where do PDEs come from?
- For a conserved quantity:

$$\begin{matrix}\text{rate of change} \\ \text{in domain}\end{matrix} \implies \begin{matrix}\text{net flow into domain} \\ + \text{ creation in domain}\end{matrix}$$

e.g. Animal population in a fixed region

<u>Quantitatively</u>

- Let $u(x,t)$ be density of some quantity in 1d domain.
- The amount of quantity in section dx is $u(x,t)Adx$.
- Let $\phi(x,t)$ be the flux, which is the rate of quantity crossing section x at time t. Note that $\phi(x,t)dt$ is the amount of stuff, and so is $u(x,t)dx$.
- $A\phi(x,t)$ is amount of quantity passing x at time t
- Let $f(x,t)$ be the rate at which quantity is created.
- $f(x,t)Adx$ is amount of quantity created in width dx per unit time.

<u>Remarks</u>:

- $\phi(x,t) > 0 \implies$ flux is L to R
- $\phi(x,t) < 0 \implies$ flux is R to L
- $f(x,t) > 0$ is source at x (creation)
- $f(x,t) < 0$ is sink at x (destroyed)

<u>Balance</u>:

$$\begin{matrix}\text{change in}\\ \text{quantity in}\\ \text{section}[a,b]\end{matrix} = \text{flux} + \text{creation}$$

$$\begin{aligned}
\Rightarrow \frac{d}{dt}\int_a^b u(x,t)Adx &= A\phi(a,t) - A\phi(b,t) + \int_a^b f(x,t)Adx\\
\Rightarrow \int_a^b u_t Adx &= -\int_a^b \phi_x Adx + \int_a^b fAdx\\
\Rightarrow \int_a^b (u_t + \phi_x - f)dx = 0 &
\end{aligned}$$

This is true for any domain $[a,b]$

$$u_t + \phi = f \text{ where}$$

$$\begin{matrix}\text{change in}\\ \text{quantity}\end{matrix} = \text{net flux} + \text{net creation}$$

<u>Remarks</u>:
Can get a PDE for u by

- Specifying $f(x,t)$
- Choosing a relationship between ϕ and u, u_x, etc. This is called constitutive law.
- E.g. $\phi = cu \to$ advection.

<u>Diffusion</u>:

- Conservation law with no sources

$$u_t + \phi_x = 0$$

- Suppose $u(x,t)$ is density of a gas of particles. Then
  - Particles move from high u to low u.

  - Flux increase with density gradient.
  - Fick's law:

$$\phi(x,t) = -Du_x$$

    where $D > 0$ is diffusion constant.
  - This gives the diffusion equation

$$u_t = Du_{xx}$$

    which is the same as the heat equation.

Remark:

- Can combine diffusion with other effects.
  e.g. Advection-diffusion eq.

$$\text{flux: } \phi = cu - Du_x$$

  Balance:

$$\begin{aligned} u_t + \phi_x &= 0 \\ u_t + cu_x &= Du_x x \end{aligned}$$

1.4 Lecture 4 - Diffusion Equation

Recap:

- Conservation Laws (Logan 1.2)
- Density: $u(x,t)$
- Flux: $\phi(x,t)$ (movement of u)
- Sources: $f(x,t)$ (creation of u)
- Balance in domain $[a,b]$

$$\frac{d}{dt}\int_a^b Au(x,t)dx = A\phi(a,t) - A\phi(b,t) + \int_a^b Af(x,t)dx$$

$$\int_a^b u_t dx = -\int_a^b \phi_x dx + \int_a^b f dx$$

$$\int_a^b (u_t + \phi_x - f)dx = 0$$

This holds for any $[a, b]$

$$u_t + \phi_x = f$$

$$\begin{matrix}\text{change in} \\ \text{quantity}\end{matrix} + \text{flux} = \text{source}$$

Diffusion equation.
Suppose $u(x,t)$ is density/concentration of substance. We have Fick's Law for flux:

$$\phi(x,t) = -Du_x,\ D > 0$$

- Movement from high u to low u
- Steeper gradient gives higher flux

Conservation/balance Law:

$$\begin{aligned} u_t + \phi_x &= 0 \\ \Rightarrow u_t &= Du_{xx} \end{aligned} \qquad \text{(Diffusion/heat eq)}$$

Remarks:

- D is called diffusivity or thermal conductivity.
- Some physical settings have a non-constant diffusivity $D(x)$

  $$\phi = -D(x)u_x$$

  This yields the PDE

  $$\begin{aligned} u_t &= (Du_x)_x \\ &= D'u_x + Du_{xx} \end{aligned}$$
  $$\text{where } D' = \frac{dD}{dx}$$

- Can also find nonlinear diffusion equations where $D = D(u)$

  $$u_t = (D(u)u_x)_x$$

- We can combine diffusion with other terms, e.g. diffusion-advection

$$u_t + cu_x = Du_{xx}$$
$$\phi(x,t) = -Du_x + cu$$

Dimensions/units:
Suppose $u(x,t)$ is temperature, then

$$[u] = \text{temp}$$
$$[u_t] = \frac{\text{temp}}{\text{time}}$$
$$[u_{xx}] = \frac{\text{temp}}{\text{length}^2}$$

So $u_t = Du_{xx}$

$$\Rightarrow [D] = \frac{\text{length}^2}{\text{time}}$$

Consider heat flow in a region of length L. The time-scale T for temperature change is

$$T = \frac{L^2}{D} \text{ so } \ D = \frac{L^2}{T}$$

Boundary & Initial Conditions. Diffusion eqs usually have BCs and ICs. Consider heat flow in 1d with domain $0 < x < l$.

IC has form

$$u(x,0) = u_0(x)$$

where $u_0(x)$ is initial temperature profile.

There are 3 types of common BCs.

1. Dirichlet BCs
   Specify u at the boundary.

$$u(0,t) = g_1(t)$$
$$u(l,t) = g_2(t)$$

   where g_1, g_2 are given i.e. u is known at boundary.

2. Neumann BCs
   Specify u_x at the boundary.

$$u_x(0,t) = h_1(t)$$
$$u_x(l,t) = h_2(t)$$

   where h_1, h_2 are given, i.e. flux is known at boundary.

3. Robin BCs
   Specify u, u_x given at boundary.

$$au(0,t) + bu_x(0,t) = \psi_1(t)$$
$$au(l,t) + bu_x(l,t) = \psi_2(t)$$

   where ψ_1, ψ_2 are given. E.g. Newton's Law of Cooling states that heat flow $\propto$ temp difference between object and environment.

$$bu_x(0,t) = -au(0,t) + \psi_1(t)$$
$$\text{flux} = \text{object} + \text{environment}$$

<u>Remarks</u>:

- A well-posed problem is a PDE with ICs and BCs s.t. it has a unique solution with a correct physical meaning (stability).
- The above terminology for BCs applies to general PDEs.

Steady-state solutions. Many time-dependent PDEs have solutions which approach steady states $u = u(x)$, which are time-independent. E.g. diffusion/heat equation:

$$u_t = Du_{xx}$$

Steady state solution $u(x)$ has $u_t = 0$, so

$$Du'' = 0 \quad (u' = u_x)$$

This gives a 2nd-order ODE, where

$$u(x) = Ax + B$$

for constants A, B set by BCs.

$\boxed{\text{ex}}$ Consider the diffusion-decay equation

$$u_t = Du_{xx} - ru$$
$$0 < x < L,\ t, D, r > 0$$

u has the flux $\phi = -Du_x$ and is destroyed at rate ru. Take BCs:

$$u(0,t) = 0$$
$$-Du_x(L,t) = -1$$
$$t > 0$$

and IC

$$u(x,0) = g(x)$$
$$0 < x < L$$

Remarks:
This model could describe a diffusing pollutant (Du_{xx}) which decays at rate ru.

$$\begin{aligned} \text{At } 0 : & \ u(0,t) = 0 \\ & \Rightarrow u \text{ is held at } 0 \\ \text{At } L : & - Du_x(L,t) = -1 \\ & \Rightarrow \text{constant flux of } u \text{ into domain} \end{aligned}$$

Steady state problem:

$$u = u(x),\ \text{so } u_t = 0$$

BVP:

$$Du'' - ru = 0$$
$$0 < x < L$$

BCs: $u(0) = 0$, $u'(L) = \frac{1}{D}$
Here we ignore IC and assume transients have decayed.
Solution of BVP:

$$u'' = \frac{r}{D}u$$
$$u(0) = 0,\ u'(L) = \frac{1}{D}$$

General solution:

$$u = A\sinh\left(x\sqrt{\frac{r}{D}}\right) + B\cosh\left(x\sqrt{\frac{r}{D}}\right)$$
$$\text{consts } A, B$$

BC1:

$$\begin{aligned} u(0) &= 0 \\ \Rightarrow B &= 0 \\ \Rightarrow u(x) &= A\sinh\left(x\sqrt{\frac{r}{D}}\right) \end{aligned}$$

BC2:

$$\begin{aligned} u'(L) &= \frac{1}{D} \\ \Rightarrow A\sqrt{\frac{r}{D}}\cosh\left(L\sqrt{\frac{r}{D}}\right) &= \frac{1}{D} \\ \Rightarrow u(x) &= \frac{\sinh\left(x\sqrt{\frac{r}{D}}\right)}{\sqrt{rD}\cosh\left(L\sqrt{\frac{r}{D}}\right)} \end{aligned}$$

<u>Remarks</u>:

- An ODE BVP may not have a solution. Physically the PDE may not have a steady state solution.
- The steady state solution could be unstable in general. (Here it turns out that $\lim_{t\to\infty} u(x,t) \to$ steady state.)

Summary. Diffusion in 1d, for $u(x,t)$

$$\begin{aligned} u_t &= Du_{xx} \\ D &> 0 \end{aligned}$$

Can consider higher dimensions for $u(x,y,t)$

<u>Preview</u>: Vibrations and waves
Consider the 1d wave equation for $u(x,t)$:

$$u_{tt} = C^2 u_{xx}$$

This describes, for example, waves on a string. C has dimensions length/time and is the speed of waves. 2nd-order in time $\to$ waves (contrasts with 1st-order in time $\to$ diffusion).

We have the string fixed at length $x = 0, L$ and $u(x,t)$ is the height of the string at time t. And we have the following equation of motion for the string

$$u_{tt} = \sqrt{\frac{T}{\rho}} u_{xx}$$

where T is string tension, and ρ is the density of string.

1.5 Lecture 5 - Wave Equation

Recap:

- Diffusion: $u_t = Du_{xx}$, $D > 0$
- Wave: $u_{tt} = c^2 u_{xx}$, c is speed

Vibrations of a string.

- String fixed at $x = 0, x = l$.
- $u(x,t)$ is the height of the string at time t, position x
- Density $\rho = \frac{\text{mass}}{\text{length}}$

Assumptions:

- Small deflections, i.e. $|u_x|$ is small.
- Only force acting on string is tension (no gravity, damping, etc.)
- No horizontal motion of string (only vertical)

Let $T(x,t)$ be the tension, T acts tangent to the string.
Let $\theta(x,t)$ be the angle of the tangent to the string, so

$$\cos\theta = \frac{1}{\sqrt{1+u_x^2}} \qquad \sin\theta = \frac{u_x}{\sqrt{1+u_x^2}}$$

So T is the tension force along the string. Therefore, total force on the string in region $[a, b]$ is

$$\text{horiz:} \quad T\cos\theta\Big|_a^b = \frac{T}{\sqrt{1+u_x^2}}\Big|_a^b$$

$$\text{vert:} \quad T\sin\theta\Big|_a^b = \frac{Tu_x}{\sqrt{1+u_x^2}}\Big|_a^b$$

We can simplify this using the small deflection assumption, $|u_x| \ll 1$

$$\begin{aligned}\text{Taylor series} : (1+u_x^2)^{\frac{1}{2}} &= 1 + \frac{1}{2}u_x^2 - \frac{1}{8}u_x^4 + \dots \\ &\approx 1\end{aligned}$$

where each u_x term gets smaller.

So the forces become

$$\begin{aligned}\text{horiz} : \quad & T\cos\theta\Big|_a^b \approx T\Big|_a^b = \int_a^b T_x dx \\ \text{vert} : \quad & T\sin\theta\Big|_a^b \approx Tu_x\Big|_a^b = \int_a^b (Tu_x)_x dx\end{aligned}$$

The accelerations are

$$\begin{aligned}\text{horiz} : \quad & 0 \quad \text{(by assumption)} \\ \text{vert} : \quad & \int_a^b u_{tt} dx\end{aligned}$$

Using Newton's 2nd Law, $F = ma$

$$\begin{aligned}\text{horiz} : \quad & \int_a^b T_x dx = 0 \\ \text{vert} : \quad & \int_a^b (Tu_x)_x dx = \int_a^b \rho u_{tt} dx\end{aligned}$$

This holds for all intervals $[a, b]$
$\Rightarrow T_x = 0$ so $T(x, t) = T(t)$

and $(Tu_x)_x = Tu_{xx}$

$$\text{vert} : Tu_{xx} = \rho u_{tt}$$
$$\Rightarrow u_t t = c^2 u_{xx} \text{ for } c = \sqrt{\frac{T}{\rho}}$$

Remarks:

- Typically assume $T(t)$ is also time-independent so $T = \text{const}$, and speed $c = \sqrt{\frac{T}{\rho}}$ is const.
- For constant c, the general solution is

  $$u(x,t) = F(x - ct) + G(x + ct)$$

  where F is a right-moving wave, and G is a left-moving wave
- For a plucked string, our BCs are

  $$u(0,t) = u(l,t) = 0$$

- To get a specific solution, we need 2 ICs, since the equation is 2nd order in time. E.g. we need

  $$\text{Init position} : u(x,0) = f(x)$$
  $$\text{Init velocity} : u(x,0) = g(x)$$
  $$\text{for } 0 < x < l$$

- There are many variations, e.g.:
  - Air resistance term, ru_t:

    $$u_{tt} - c^2 u_{xx} + ru_t = 0$$

  - External force, $f(x,t)$:

    $$u_{tt} - c^2 u_{xx} = f(x,t)$$

Standing waves. Consider a vibrating string

$$\begin{gathered} u_{tt} = c^2 u_{xx} \\ \text{BCs} : u(0,t) = u(l,t) = 0 \end{gathered}$$

Standing waves are solutions of the form

$$\begin{aligned} u(x,t) = & \phi(x)\cos\omega t \\ \text{or } & \phi(x)\sin\omega t \\ \text{or in general } & \phi(x)e^{i\omega t} \end{aligned}$$

Each part of the string oscillates with:

- Same freq ω, in phase.
- Different amplitude $\phi(x)$

Solutions only exist for certain ω

$$\begin{gathered} \text{substitute } u = \phi(x)\cos\omega t \text{ into wave eq} \\ -\omega^2\phi\cos\omega t = c^2\phi''\cos\omega t \\ \text{with } \phi(0)\cos\omega t = \phi(l)\cos\omega t = 0 \ \forall\ t \end{gathered}$$

This gives us an ODE BVP:

$$\phi'' = -\frac{\omega^2}{c^2}\phi,\ \phi(0) = \phi(l) = 0$$

We have the following general solution

$$\phi(x) = A\cos\frac{\omega x}{c} + B\sin\frac{\omega x}{c}$$

BC1: $\phi(0) = A = 0$

$$\phi(x) = B\sin\frac{\omega x}{c}$$

BC2: $\phi(l) = B\sin\frac{\omega l}{c}$

$$B = 0 \text{ or } \sin\frac{\omega l}{c} = 0$$

If $B = 0$ then $\phi(x) = 0$ and $u(x,t) = 0$. We are interested in nonzero solutions so we need $\sin \frac{\omega l}{c} = 0$

Note: l and c are fixed by the problem, ω is the frequency of the standing wave. The condition $\sin \frac{\omega l}{c} = 0$ means the allowed frequencies ω of a standing wave satisfy $\sin \frac{\omega l}{c} = 0$

$$\Rightarrow \frac{\omega l}{c} = n\pi \text{ for } n \in \mathbb{Z}$$

Allowed frequencies: $\omega_n = \frac{n\pi c}{l}$
Modes: $\phi_n(x) = \sin \omega_n x = \sin\left(\frac{n\pi x}{l}\right),\ n \in \mathbb{Z}$

- $B\phi_n(x)$ satisfies the BVP for any constant B, modes defined up to a multiplicative constant.
- $n = 0$ gives $\phi_n = 0$ so it is trivial.
- $n < 0$ gives the same mode as $n > 0$ up to a sign: $\phi_n(x) = -\phi_{-n}(x)$
- We can take the modes to have $n > 0$

$$\phi_n(x) = \sin\left(\frac{n\pi x}{l}\right),\ n = 1, 2, 3, \ldots$$

The standing waves are

$$\begin{aligned} u_n(x,t) &= \cos\left(\frac{n\pi ct}{l}\right)\sin\left(\frac{n\pi x}{l}\right) \\ &= \cos(\omega_n t)\phi_n(x),\ n = 1, 2, 3, \ldots \end{aligned}$$

Remarks:

- The BVP problem is an eigenvalue problem. The frequencies ω_n are the c-values and modes ϕ_n are eigenfunctions.
- Eigenfunctions ϕ_n are defined up to a constant multiplier, like eigenvectors.
- The frequencies ω_n are the fundamental frequencies of the string.

Quantum Mechanics (Logan 1.6)
Consider a particle in 1d. Classically, the particle has a position $x \in \mathbb{R}$. In quantum mechanics, the state of the particle is a function $\Psi(x,t) \in \mathbb{C}$ called the wave function. The probability density of finding the particle at x is $|\Psi(x,t)|^2$. Ψ obeys the Schrödinger equation (SE):

$$i\hbar\Psi_t = -\frac{\hbar^2}{2m}\Psi_{xx} \quad \text{(free particle)}$$

where $\hbar$ is Planck's constant
m is particle mass

If the 'energy cost' of being at position x is $V(x)$, then the SE is

$$i\hbar\Psi_t = -\frac{\hbar^2}{2m}\Psi_{xx} + V(x)\Psi$$

Remarks:

- The particle is typically more likely to be at places where $V(x)$ is lower.
- Probability is conserved

$$\int_{-\infty}^{\infty} |\Psi|^2 dx = 1 \ \forall t$$

- Looking for solutions of the form

$$\Psi(x,t) = e^{-iEt/\hbar} \cdot \phi(x)$$

  gives the following time=independent SE:

$$E\phi = -\frac{\hbar^2}{2m}\phi'' + V(x)\phi$$

  Where E has the interpretation of 'energy'

For a particle in a '1d box,' $0 < x < l$, we have

$$E\phi = -\frac{\hbar^2}{2m}\phi'', \ 0 < x < l$$

with the following BCs

$$\phi(0) = \phi(l) = 0$$

$\Rightarrow$ standing wave solutions

1.6 Lecture 6 - Higher Dimensional PDEs

Recap:

- 1st order PDEs, characteristics, change of coords.
- 2nd order PDEs (1 space dimension), diffusion/heat eq, with advection, decay, etc.

$$u_t = Du_{xx}$$

- Steady state solutions
- Wave equation

$$u_{tt} = c^2 u_{xx}$$

- Standing waves, with BCs and ICs.
- Dirichlet (u is known on boundary), and Neumann (u_x is known on boundary).
- ODEs: separation of variables, homogeneous solutions, 2nd order constant coefficients.

Higher dimensional PDEs (Logan 1.7)

- Most physical processes occur in more than 1d.
- We can generalize 1d derivations using vector calculus.

Heat flow in 2d/3d. 2d and 3d are similar, so we work in 3d. Consider heat flow in domain $\Omega \subseteq \mathbb{R}^3$.

- Let $u(x, y, z, t)$ be the temperature at point $(x, y, z) \in \Omega$ at time t.
- Let Cu be the thermal energy density.
- Let $\Phi(x, y, z, t)$ be the energy flux vector, i.e. Φ is the vector field describing heat flow.
- The rate of heat flow across a surface element, dS is $\Phi \cdot dS$

- Recall that $dS = n \cdot dA$, where n is the normal vector, and dA is the area element.
- In 2d, flow across a line element is $\Phi \cdot ndl$
- Let $f(x, y, z, t)$ be the rate at which heat is produced.
- Consider thermal energy in subdomain $B \subseteq \Omega$

Conservation/balance:

$$\frac{d}{dt}\int_B CudV = -\int_{\partial B} \Phi \cdot dS + \int_B fdV$$

$$\begin{array}{c}\text{change in}\\ \text{energy}\end{array} = \begin{array}{c}\text{flux out}\\ \partial B \text{ is boundary}\end{array} + \text{sources/sinks}$$

Remark: this applies to virtually every spatially varying density.

Using the divergence theorem, we can get the PDE.
Let $\vec{F}$ be a (smooth) vector field in region $B \subseteq \mathbb{R}^3$. Then

$$\int_{\partial B} F \cdot dS = \int_B \nabla \cdot FdB$$

where $\vec{F} = (F_1, F_2, F_3)$ and $\nabla \cdot F = \partial_x F_1 + \partial_y F_2 + \partial_z F_3$

Apply this to conservation equation:

$$\frac{d}{dt}\int_B CudV = -\int_{\partial B} \Phi \cdot dS + \int_B fdV$$
$$\int_B Cu_t dV = -\int_B \nabla \cdot \Phi dV + \int_B fdV$$

This is true for any $B \subseteq \Omega$

$$\Rightarrow Cu_t = -\nabla \cdot \Phi + f$$

To get a PDE for u, we need a relationship between Φ and u. In 1 space dimension, we had $u = u(x, t)$, $\phi(x, t)$, and $\phi = -Ku_x$ with constant $K > 0$

(in 1d, ϕ is effectively a scalar).

Fick's Law in higher dimensions:

$$\begin{aligned} \Phi &= -K\nabla u \quad \text{(constant } K > 0) \\ \nabla u &= (u_x, u_y, u_z) \end{aligned}$$

This gives the PDE

$$\begin{aligned} Cu_t &= -\nabla \cdot \Phi + f \\ &= K\nabla \cdot (\nabla u) \end{aligned}$$

where

$$\begin{aligned} \nabla \cdot (\nabla u) &= \nabla \cdot (u_x, u_y, u_z) \\ &= u_{xx} + u_{yy} + u_{zz} \\ &:= \nabla^2 u \quad \text{(Laplacian of } u) \end{aligned}$$

This gives the PDE

$$\begin{gathered} Cu_t = K\nabla^2 u + f \\ \text{for } (x, y, z) \in \Omega \end{gathered}$$

<u>Definition</u>: The Laplacian
Let u be a scalar function which depends on space and possibly time. The Laplacian of u, $\nabla^2 u$ is the sum of 2nd derivatives with respect to space (in Cartesian coords).

<u>Remarks</u>:

- The Laplacian ∇^2 is a linear differential operator.
- We've only defined ∇^2 for scalar functions.
- Laplacian of u is sometimes written as Δu

$$\text{e.g. } \Delta u = \nabla^2 u = \nabla \cdot (\nabla u)$$

<u>Heat/diffusion eq in higher dimensions</u>.

$$u_t = D\nabla^2 u, \text{ constant } D > 0$$

for $u = u(x, y, z, t)$ and $(x, y, z) \in \Omega \subseteq \mathbb{R}^3$

Heat/diffusion eq with heat source.

$$u_t = D\nabla^2 u + f$$

for $f = f(x, y, z, t)$

Remarks:

- The time-independent cases also give important equations
  Poisson equation:

$$\begin{aligned} \nabla^2 u = -f \quad &\in \Omega \\ u = u(x, y, z) \end{aligned}$$

  Laplace equation:

$$\begin{aligned} \nabla^2 u = 0 \quad &\in \Omega \\ u = u(x, y, z) \end{aligned}$$

  Wave equation:

$$u_t t = c^2 \nabla^2 u \quad \in \Omega$$

- These equations are fundamental for describing spatially varying quantities in 2d, 3d. E.g. fluid flow, vibrations, quantum mechanics, electromagnetism.

Boundary Conditions. Just like in 1d, PDEs in higher dimensions need BCs. Consider the heat equation $u_t = D\nabla^2 u$ in 1d, where $\Omega = [0, L] \subseteq \mathbb{R}$

- Dirichlet BCs: $\begin{cases} u(0, t) = g_1(t) \\ u(L, t) = g_2(t) \end{cases}$ i.e. u is known on boundary.

- Neumann BCs: $\begin{cases} u_x(0, t) = g_1(t) \\ u_x(L, t) = g_2(t) \end{cases}$ i.e. u_x is known on boundary (flux).

In 2d, we have the area $\Omega \in \mathbb{R}^2$ and $\partial\Omega$ is the boundary (curve).

- Dirichlet BC: $u(x, y, t) = g(x, y, t)$ for $(x, y) \in \partial\Omega$ (boundary).
  This can be written as
  $$u\big|_{\partial\Omega} = g$$
  i.e. u is known on boundary.

- Neumann BC: $n(x, y) \cdot \nabla u(x, y) = g(x, y, t)$ for $(x, y) \in \partial\Omega$ where n is the unit normal to $\partial\Omega$ at (x, y).
  This can be written as
  $$\vec{n} \cdot \nabla u\big|_{\partial\Omega} = g$$
  i.e. flux out of domain $n \cdot \nabla u$ is known.

$\boxed{\text{ex}}$ Vibrating drum.
A drum occupies region $\Omega \subseteq \mathbb{R}^2$, $u(x, y, t)$ is height of the drum, and the drum is clamped on its boundary.

$$u\big|_{\partial\Omega} = 0 \quad \text{(Dirichlet BC)}$$

The height of the drum obeys

$$\begin{aligned} u_{tt} &= c^2 \nabla^2 u \ \text{ on } \Omega \\ u\big|_{\partial\Omega} &= 0 \end{aligned}$$

where $c^2 = \frac{T}{\rho} = \frac{\text{tension}}{\text{density}}$

Standing wave solutions:

$$\nabla^2 u = -\frac{\omega^2}{c^2} u, \quad u\big|_{\partial\Omega} = 0$$